

ENV 5128

Smart Air Quality Monitoring & Air Pollution Control

Today's Topics

- Refresh of air pollution basics
- Expectation
 - Understand mechanisms of bad ozone formation
 - Understand sources of acid deposition and trend
 - Understand mechanisms of O₃ layer depletion
 - Understand size distributions & sources of PM

Tropospheric ozone formation

LA Smog



Beijing smog

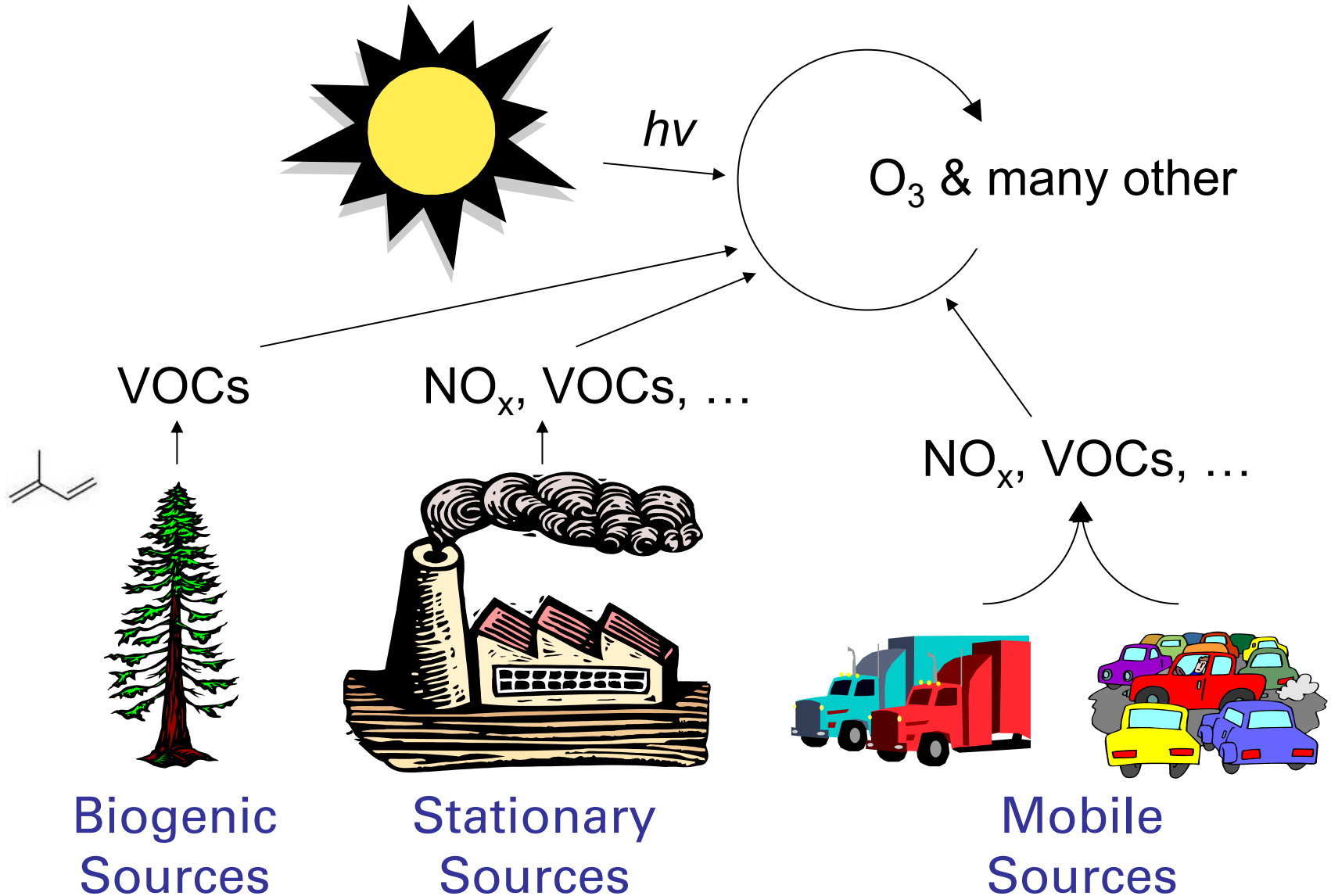


Smog in 20 minutes. Beijing, China. Jan 2nd, 2017
Credit: YouTube/Chas Pope

Smog

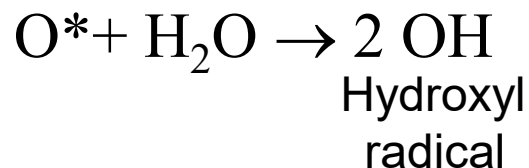
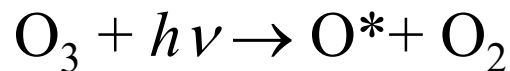
- Photochemical Smog (smoke+fog)
 - ‘Soup’ of gases and aerosols (primary + secondary)
 - Hazardous to health
 - Damage to materials, crops, visibility degradation, effects on climate and clouds
- There are many chemicals within smog
 - PM, O₃, HCs, NO_x, SO₂ etc
 - Ozone is a major component

photochemical smog



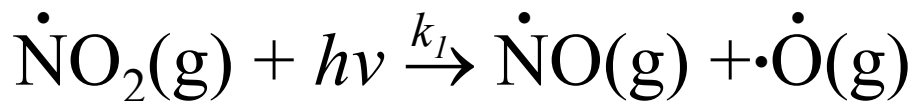
Background ozone chemistry

- O_3 do naturally occur in troposphere
- Approximately 10% of O_3 is in the troposphere
 - The rest is in stratospheric ozone layer
- Photolysis of O_3 leads to the production of hydroxyl radical (OH)
 - Tropospheric O_3 is important source
- Reaction with OH radical is the most important loss mechanism for almost all trace gases

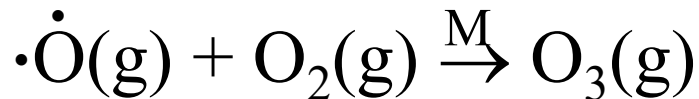


Background ozone chemistry

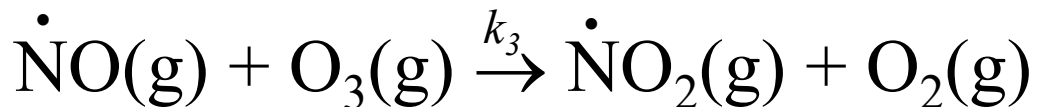
- Now consider a clean (background) atmosphere
- Three important reactions:



photolysis of NO_2
(production of O and NO)

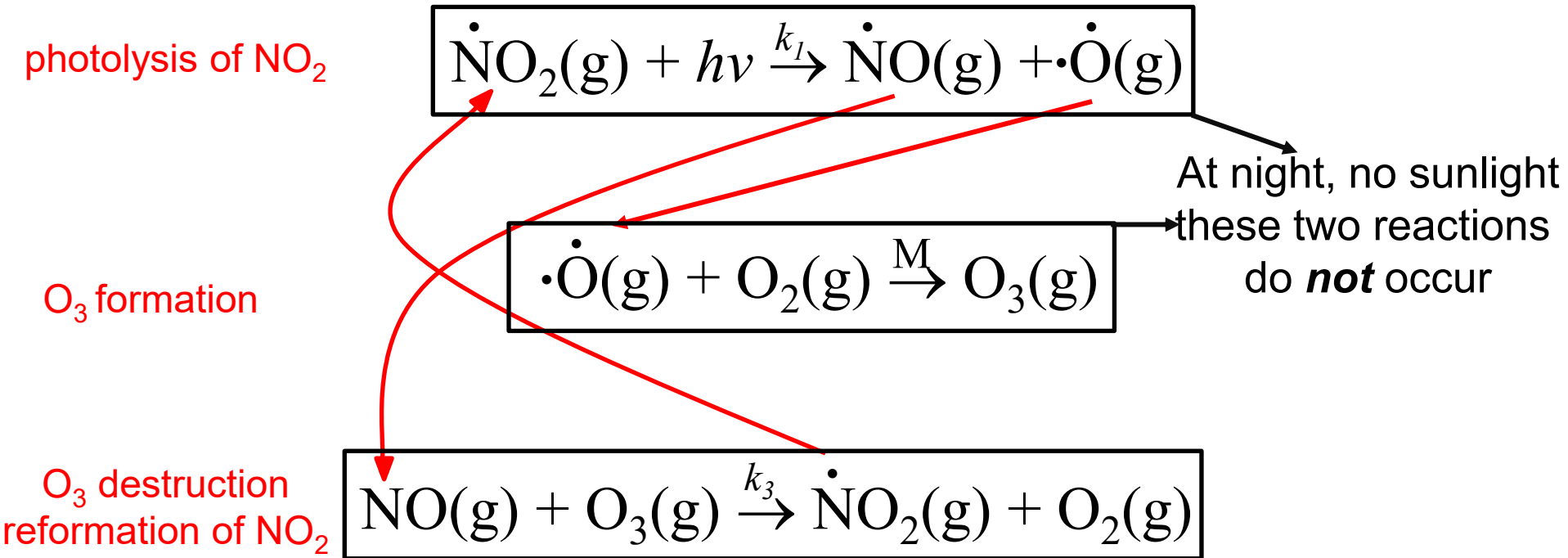


rxn of atomic oxygen with O_2 to
form O_3 (almost instantaneous)



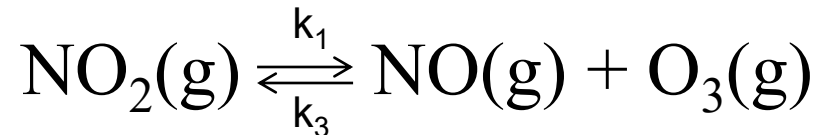
rxn of NO with O_3 (ozone
destruction, reformation of NO_2)

Background ozone chemistry



Background ozone chemistry

- A simplify representation:



- Ozone going through a cycle of formation/removal
- At equilibrium: ***photostationary state (pss)***
 - Only in a few minutes with sunlight
- At this time, ozone concentration is given by:

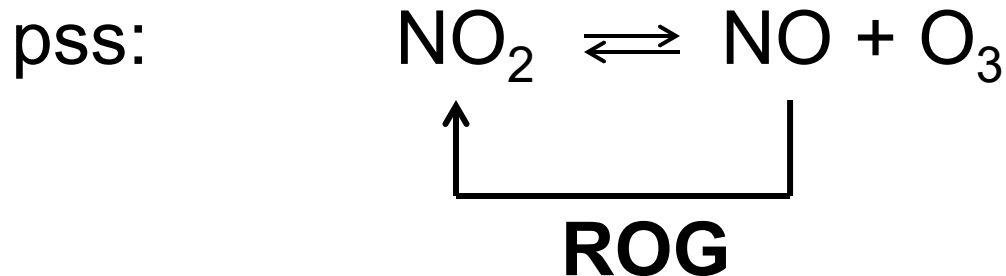
$$[\text{O}_3] = \frac{k_1 [\text{NO}_2]}{k_3 [\text{NO}]} \propto \frac{[\text{NO}_2]}{[\text{NO}]}$$

More NO₂, more ozone
More NO, less ozone

Background ozone chemistry

- Under clear sky, mid-latitude, mid afternoon conditions: $k_1/k_3 = 18$ ppbv
 - Typical value, real ratio do vary
- BUT, ozone concentration normally way higher than that estimated using k_1/k_3
- Also, ozone concentration not highest near NOx emission sources, bur rather downwind
 - Recall only a few minutes to reach pss

Reactive OrGanics (ROG)



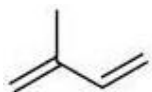
Role of reacting organics in smog formation:

- convert NO to NO₂, thereby increasing O₃ via pss
- produce aldehydes and peroxyacetyl nitrates (PANs)

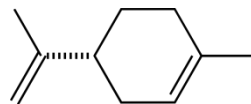
$$[O_3] = \frac{k_1 [NO_2]}{k_3 [NO]} \propto \frac{[NO_2]}{[NO]}$$

ROG sources

- Many volatile organic compounds (VOCs)
- Sources
 - anthropogenic
 - transportation sector (50% of anthropogenic)
 - industrial sector (50%; vaporization of organics)
 - biogenic (large; isoprene and monoterpenes)

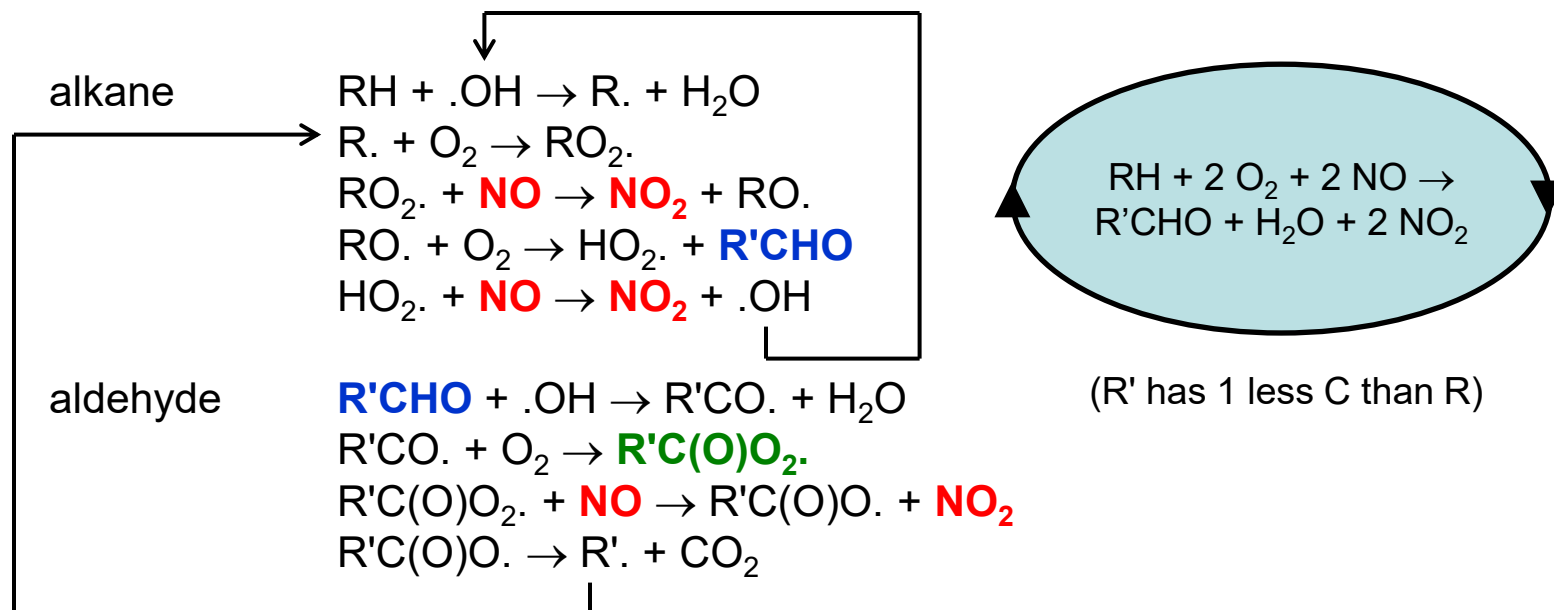


isoprene



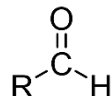
limonene

general reaction mechanism

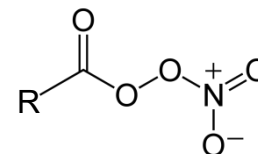


NO converted to **NO₂**

aldehydes (**R'CHO**) are formed

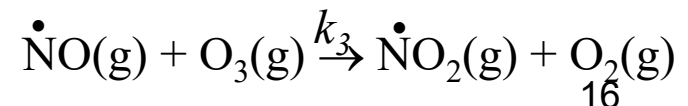
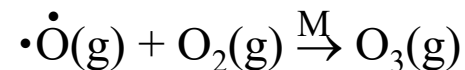
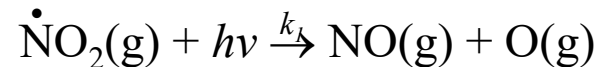
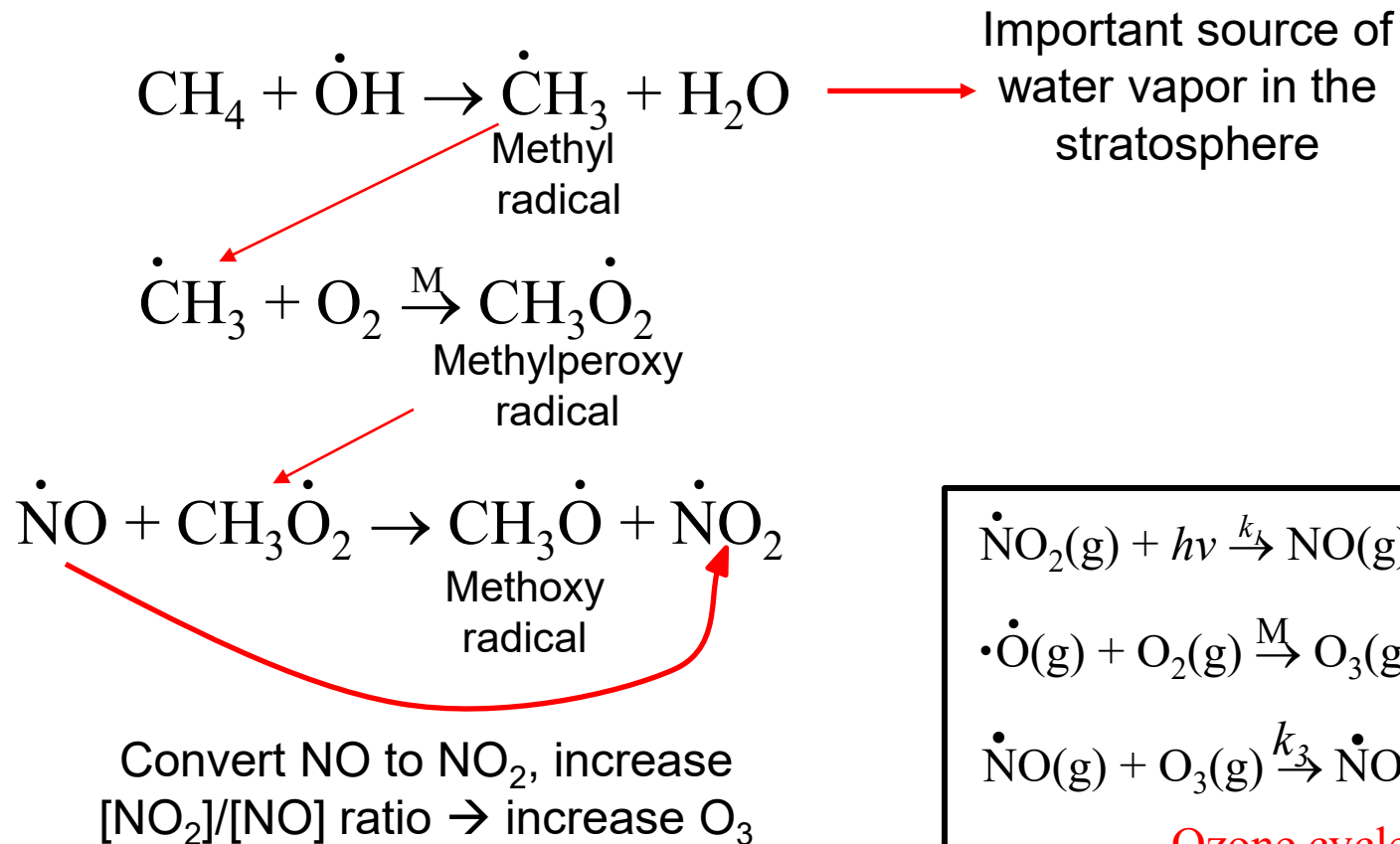


PANs (peroxyacetyl nitrates) can be formed:



Example: effect of CH₄ on O₃

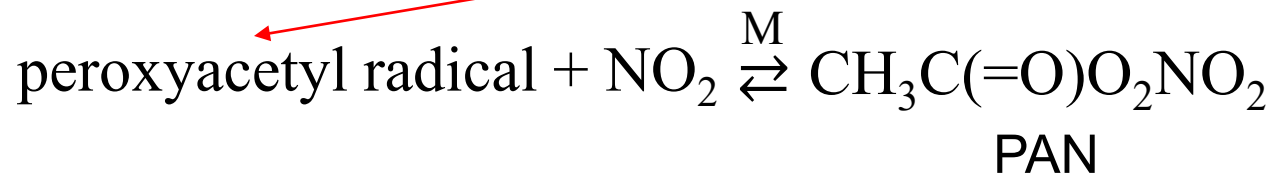
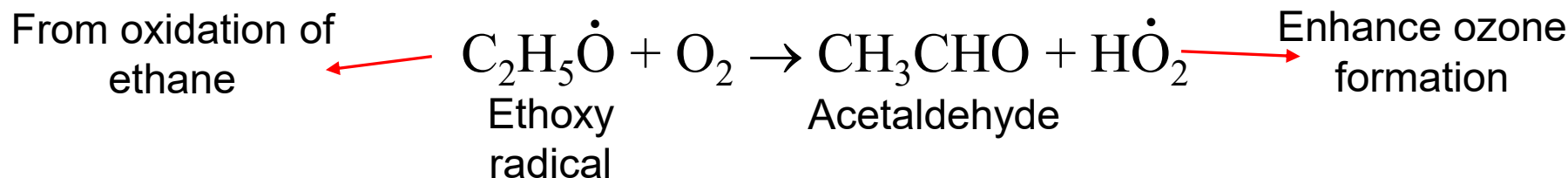
- Methane is the most abundant organic gas in the atmosphere (8-12 years of e-folding lifetime).



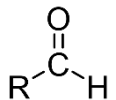
Ozone cycle

Example: PAN

- Acetaldehyde is an important product of ethane oxidation, precursor to peroxyacetyl nitrate (PAN)



relative reactivity of organics

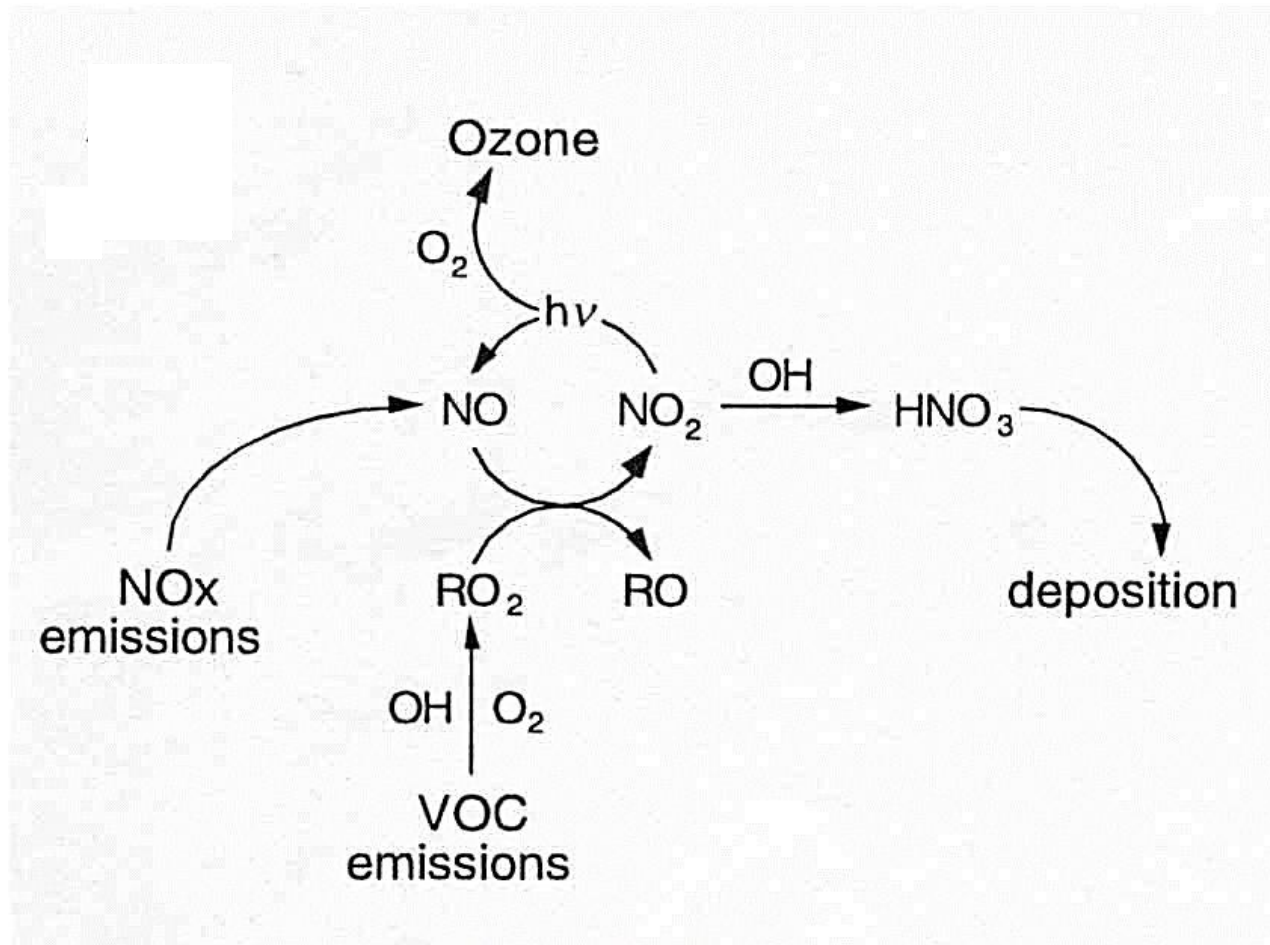
	k (25°C) cm ³ molecule ⁻¹ s ⁻¹	1/(k [.OH]) ^a
CO + .OH → CO ₂ + H.	1.5x10 ⁻¹³	36 days
CH ₄ + .OH → CH ₃ . + H ₂ O	6.3x10 ⁻¹⁵	3 years
/ C ₂ H ₆ + .OH → C ₂ H ₅ . + H ₂ O	2.6x10 ⁻¹³	30 days
∧ C ₃ H ₈ + .OH → C ₃ H ₇ . + H ₂ O	1.2x10 ⁻¹²	7 days
other alkanes, C _n H _{2n+2} + .OH → ...		6 hrs - 7 days
// C ₂ H ₄ + .OH → .CH ₂ -CH ₂ OH	8.5x10 ⁻¹²	22 hours
≡ C ₃ H ₆ + .OH → CH ₃ -.CH-CH ₂ OH	2.6x10 ⁻¹¹	7 hours
other alkenes, C _n H _{2n} + .OH → ...		2 - 8 hours
 RCHO + .OH → RC.O + H ₂ O		2 - 12 hours

^aTime takes for reactant concentration drop to 37%, assuming
[.OH] = 0.06 ppt = 1.5x10⁶ molecules/cm³

Role of ROG in O₃ formation

- ROG enhance O₃ formation by converting NO to NO₂, but it takes time for their effects to kick in
- While wind carries the mixture of NO_x and ROG away, reactions occur that increase O₃ concentration.
- Highest ozone concentration not found near NO_x emission sources (urban regions), but rather downwind
- ROG reactions produce aldehydes, many are human carcinogens

Urban ozone cycle



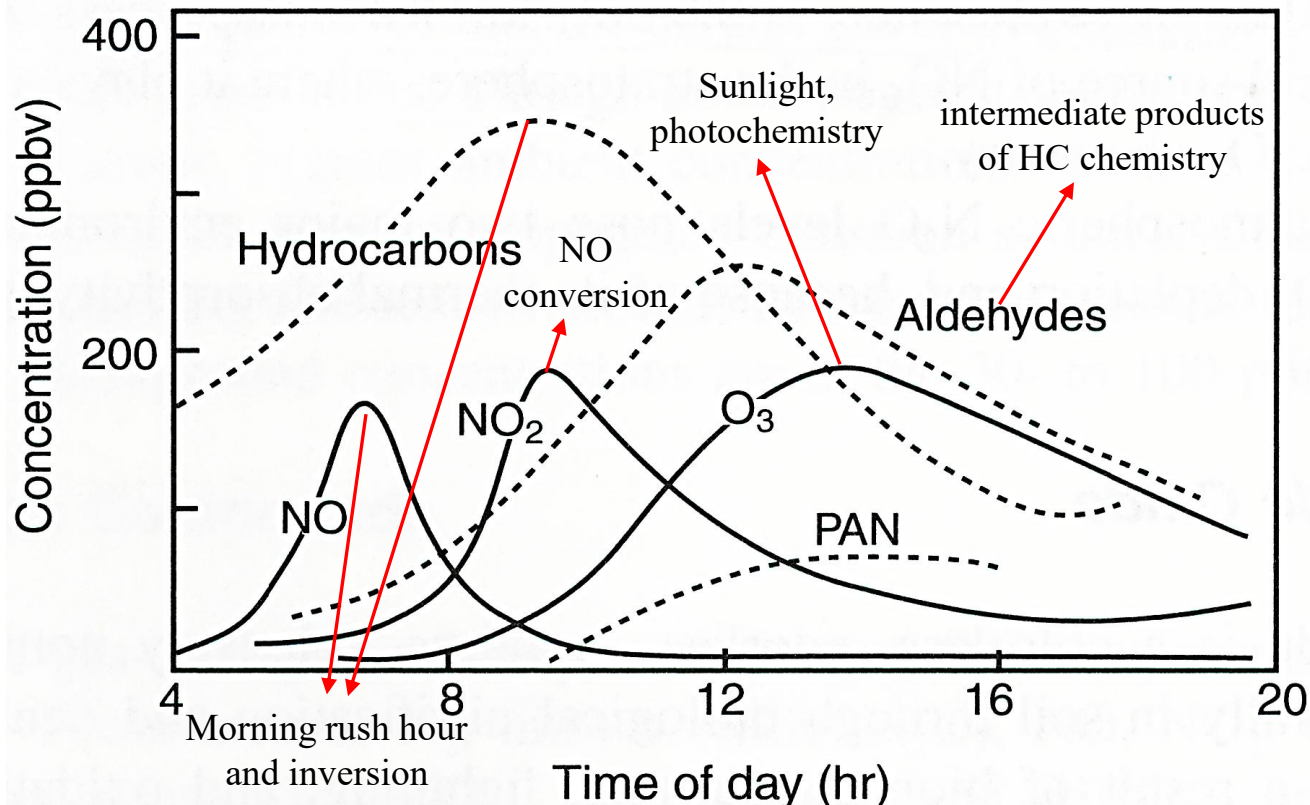
Ryerson, T. B., et al. (2001) *Science*, 292:719-722.

Urban ozone cycle

- Ozone formation potential of different VOCs differs depending on
 - Abundance of reactant
 - Rate of initial attack on precursor (by OH)
 - # of $\text{NO} \rightarrow \text{NO}_2$ conversions per VOC molecule
 - Number of cycles
 - Potential of ROG as an NO_2 sink
 - e.g. formation of PAN

Diurnal variations in smog

Three important factors: Emissions, Mixing height, Chemistry

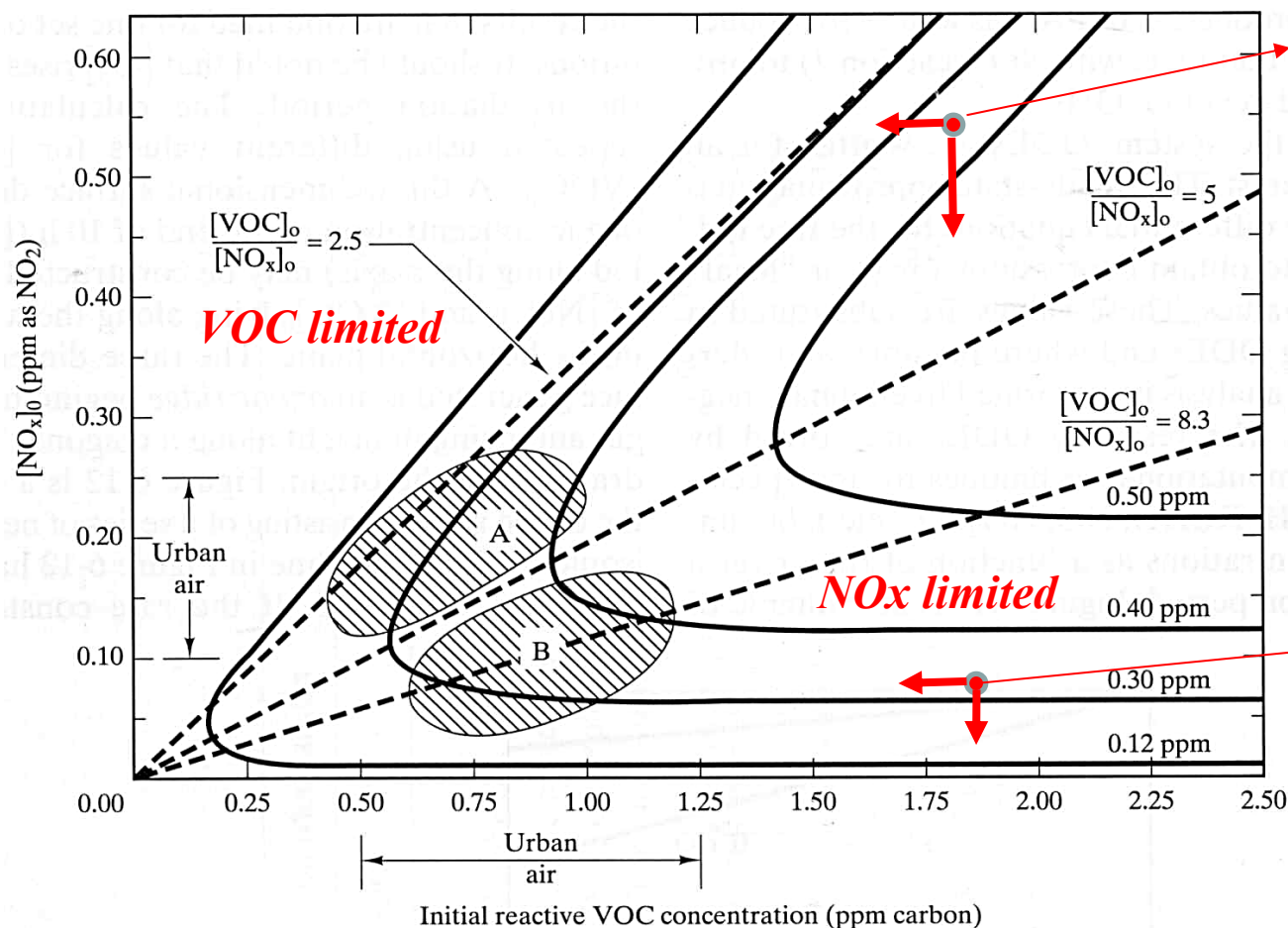


Typical pollutant concentration variation during one day (Godish, 2003)

Control of ozone

- As a secondary pollutants, ozone concentration typically reaches max at 50-100 km downwind
- Reduction in NO_x/VOC emissions does not necessarily reduce ozone concentration
- Depending on where you are at the ozone isopleth
 - Ozone concentration $\sim$ VOC & NO_x concentration

Ozone isopleth



peroxyacetyl nitrates

- Important component of photochemical smog
 - Damage plants, eye irritant, powerful tear gas
 - “Explosive” reaction
- Potential serious health effects
- Stable when cold, but quickly decomposed when hot
 - Last for long time when rises to upper atmosphere
 - Can be transported long distance (can be global)
 - Quickly decomposed if brought down near surface

Acid Deposition

Deposition

- Transfer of pollutants (and pollutant products) from the atmosphere to the earth's surface
- Have effects at many scales
 - Acidification of lakes (and soils)
 - Material degradation
 - Toxic metals deposition (e.g. Hg)
 - Persistent organic pollutants
 - Nitrogen input to ecosystems

Dry and wet deposition

- Dry deposition: Direct deposition to surfaces
- Wet deposition: Precipitation involved deposition
- Acid deposition: Creation and transport of acid constituents from atmosphere to earth surface
 - Can be both dry and wet deposition
 - Chemical reactions usually involved in both gaseous and aqueous phase

Understanding acidity

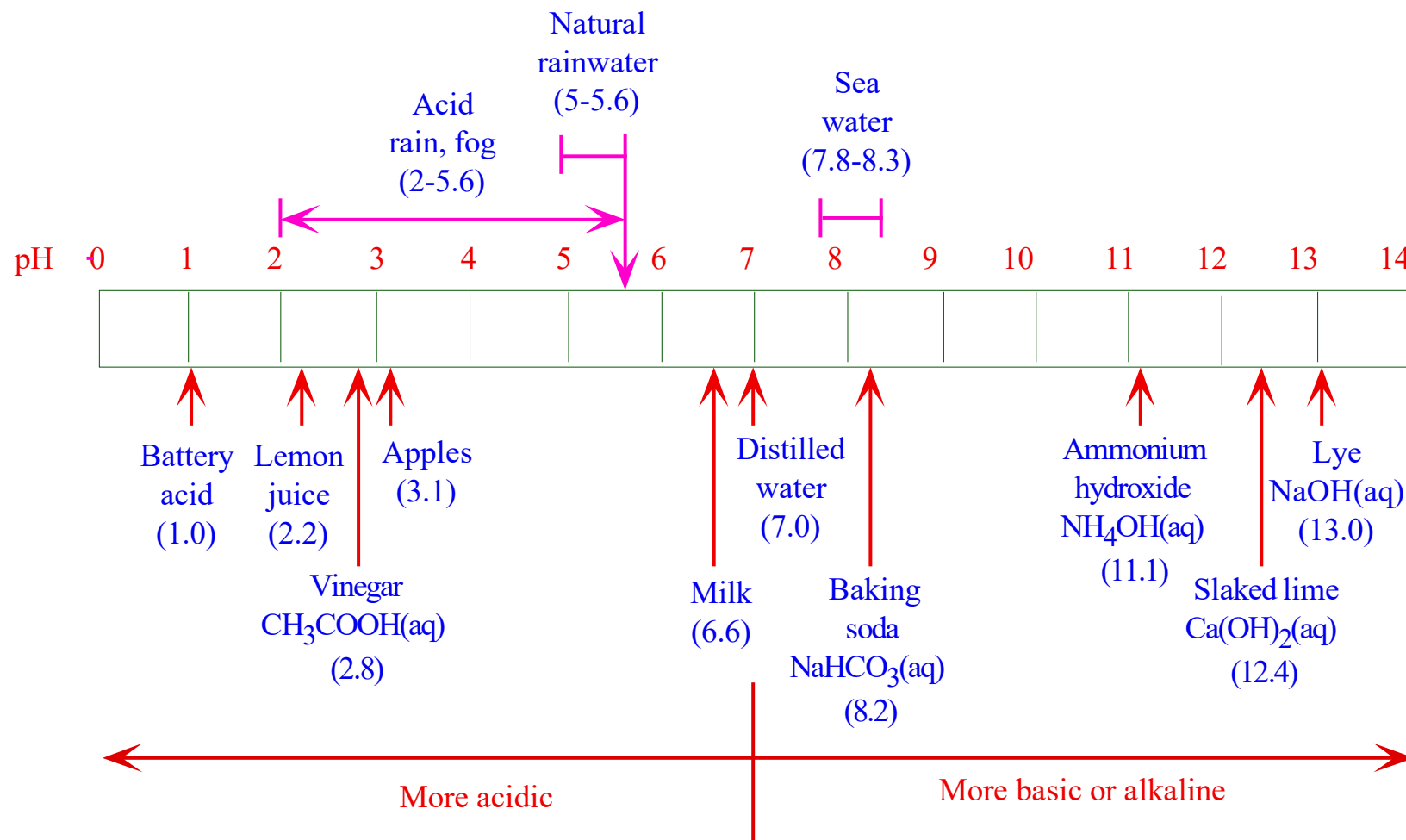
- Acidity ~ the amount of dissociated H^+ ions
 - Characterized by pH value

$$pH = -\log_{10}[H^+]$$

$[H^+]$: molarity (molar or moles/liter)

- Chemical releases $[H^+]$ ions $\rightarrow$ lower pH $\rightarrow$ increase acidity
- Chemical take up $[H^+]$ ions $\rightarrow$ higher pH $\rightarrow$ decrease acidity

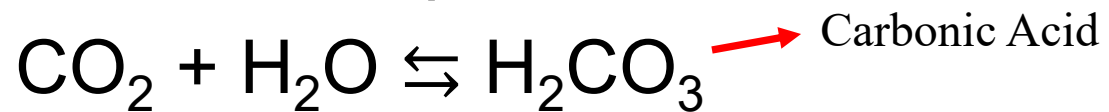
pH Scale



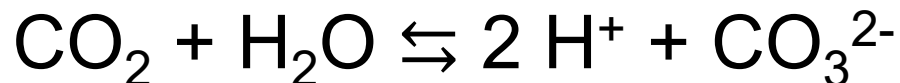
pH scale (Figure 10.3 Jacobson, 2012)

Dissociation of CO₂

- Rain water is slightly acidic naturally (pH ≈ 5.6) due to dissolved CO₂
- Acid rain: rain water with pH < 5



- Overall, we have



Main constituents of acid deposition

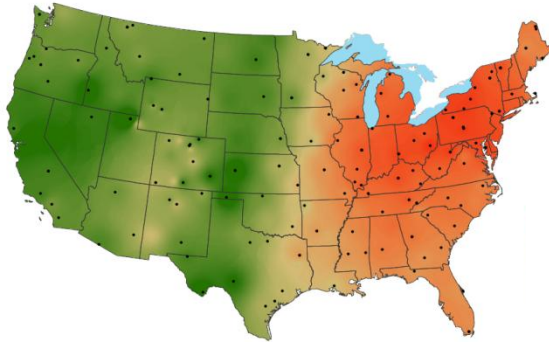
- H_2SO_4
 - Most important in eastern U.S (60 to 70 % of excess acidity)
 - From emitted SO_2
 - 70% of SO_2 emissions are anthropogenic in U.S.
- HNO_3
 - More important in western U.S.
 - Near polluted cities with fog (such as LA)
 - From emitted NO_x
 - 85% of NO_x emissions are anthropogenic in U.S

Main constituents of acid deposition

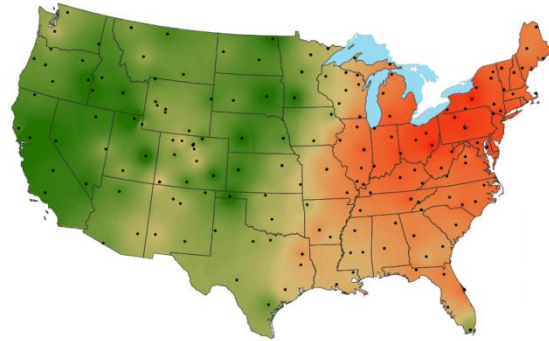
- HCl
 - Wood burning or industrial processing emissions
 - Waste incineration
 - Only about 5% of acidity
- Role of other acids are not as important
 - Formic (HCOOH) & acetic acid (vinegar, CH_3COOH)
- Acid rain control is very successful in US due to increasingly stringent emission controls

pH of rain

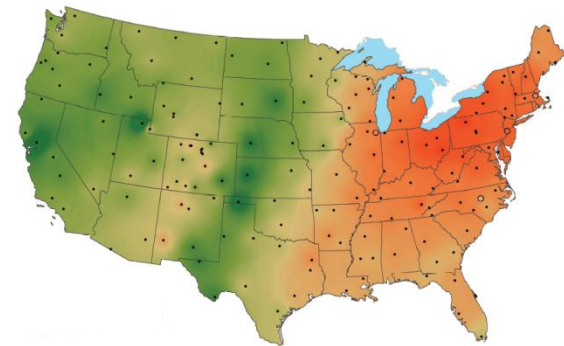
1985



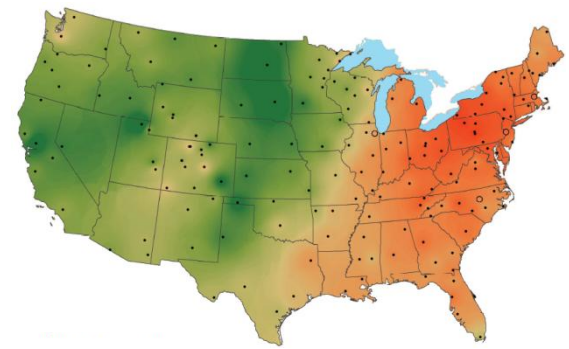
1990



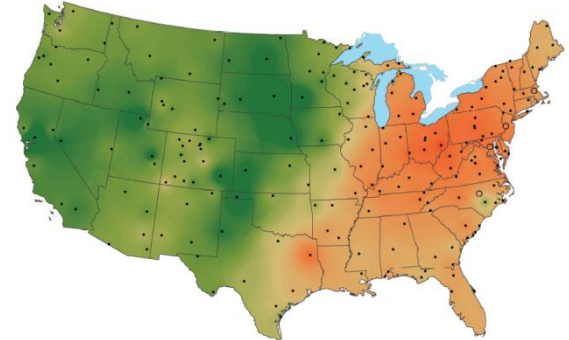
1995



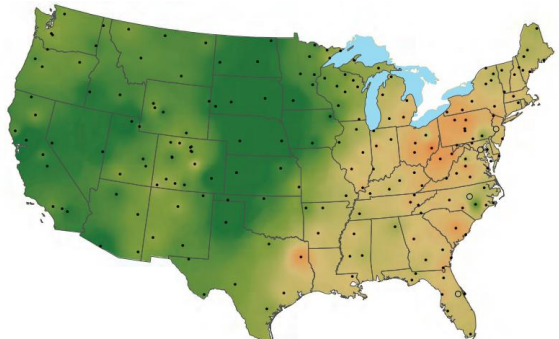
2000



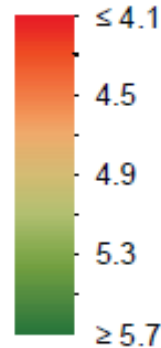
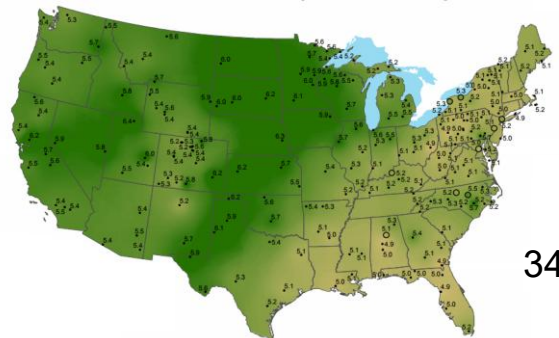
2005



2010



2015



The National Atmospheric Deposition Program (NADP) monitors precipitation chemistry, National Trends Network (NTN), <https://nadp.slh.wisc.edu/>

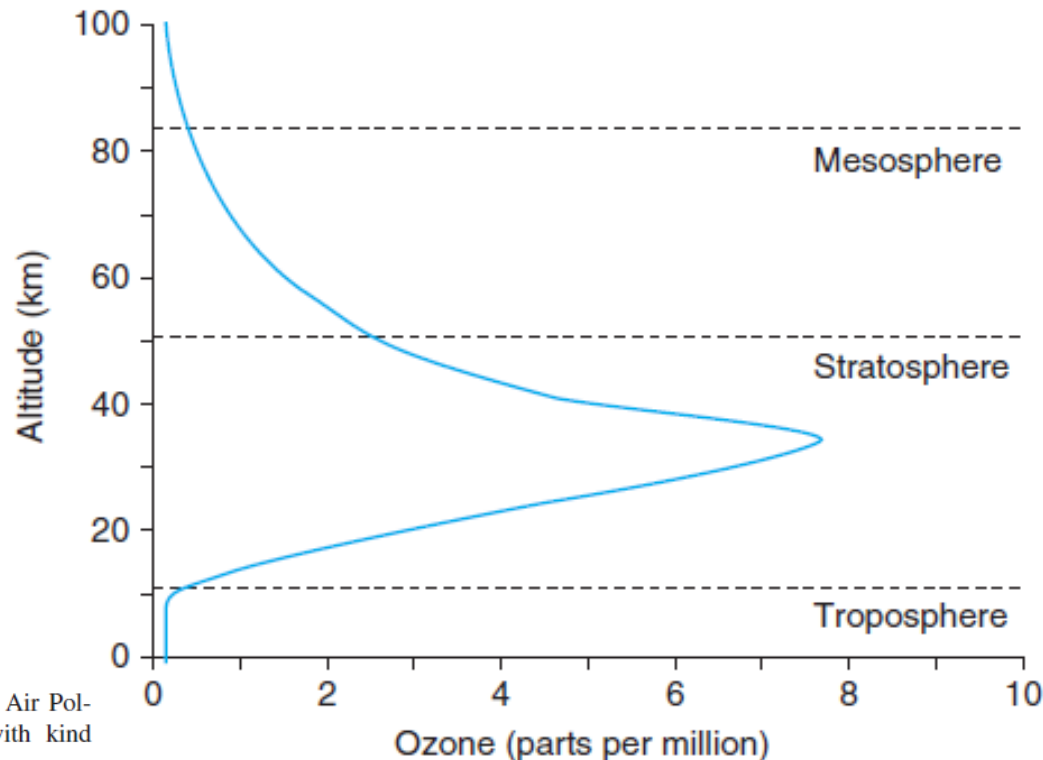
Antarctic Ozone Hole

Introduction

- Ozone presented in both troposphere and stratosphere
- In troposphere → bad ozone
 - Health effect when exposed to
 - Contribute to global warming
- In stratosphere → good ozone
 - Absorbs harmful UV
- 90% of ozone are in stratosphere

Ozone layer structure

- Vertically: peak at 25-32 km altitude
 - Sweet spot with just enough UV and O_2 to form O_3
 - Different chemistry with ground level ozone formation!



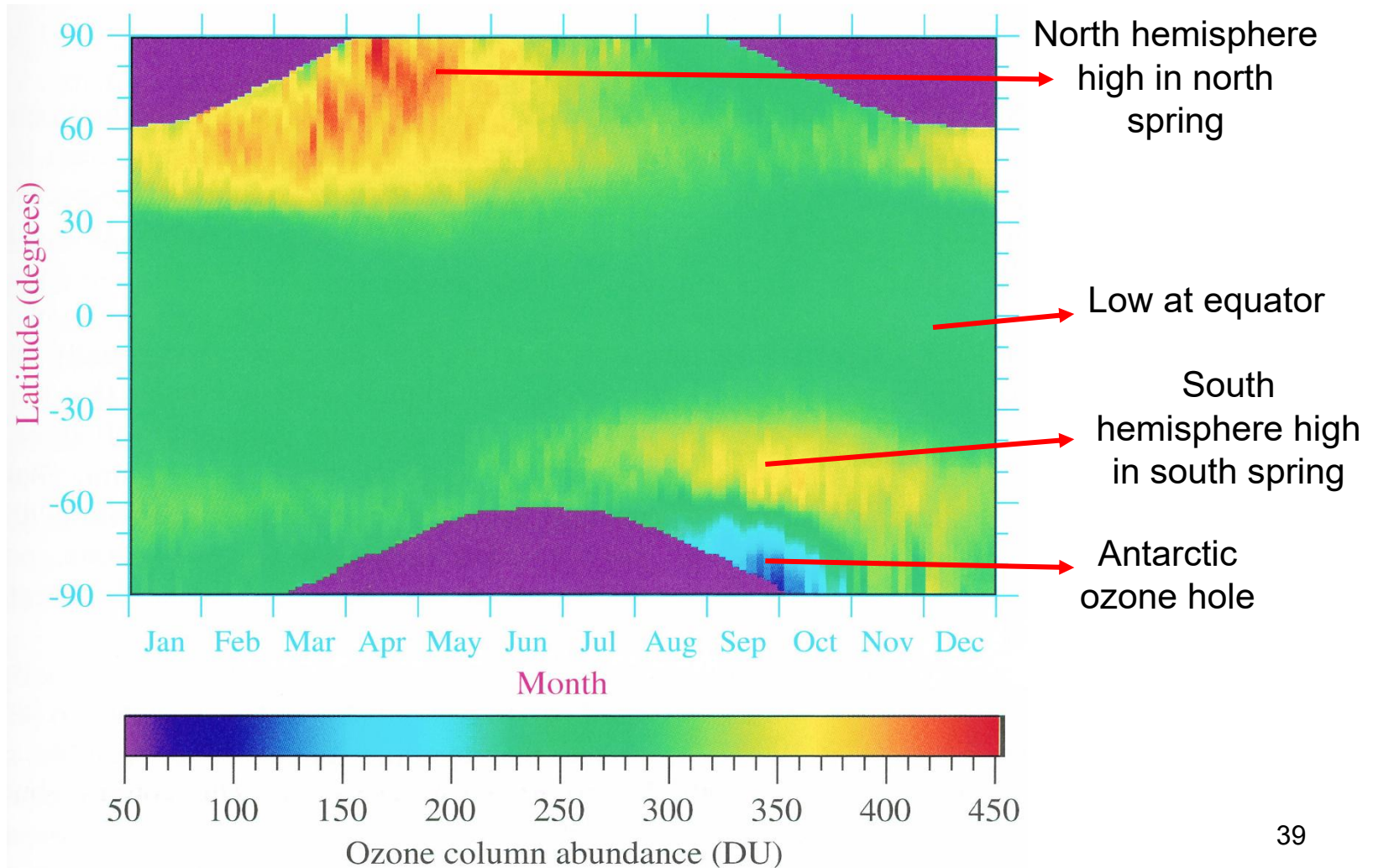
Source: The University of California Air Pollution Health Effects Laboratory, with kind permission.

Ozone layer structure

- Traditionally, ozone concentration expressed in Dubson Unit (DU)
 - Layer of gas that would be 10 μm thick under standard temperature and pressure
 - Globally ozone average concentration ~ 300 DU
- Ozone concentration varies vertically and horizontally

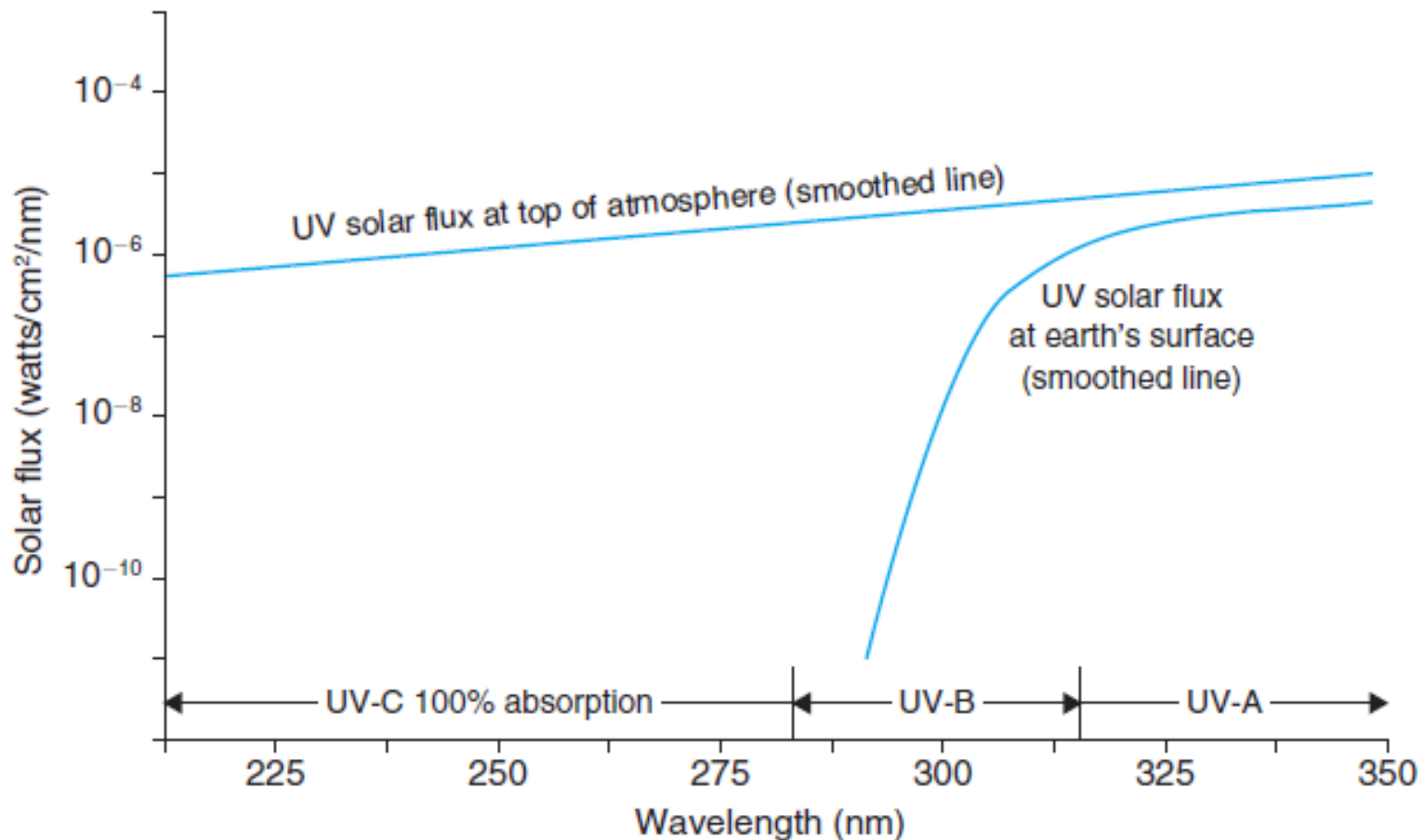
Horizontal ozone layer structure

Ozone concentration variation with time & latitude as measured by NASA TOMS satellite (Jacobson, 2002)



Ozone layer function

- Ozone absorbs harmful UV radiation



Source: The University of California Air Pollution Health Effects Laboratory, with kind permission.

Ozone layer function

- Ozone absorbs UV radiation between 250-345 nm
 - Strongly at 310 nm
 - Weakly between 310-345 nm
 - N_2 , O_2 absorbs far-UV (10-250 nm)
- Allow little UV-C, some UV-B and most UV-A reaches troposphere
- Some say, without ozone layer, no life form as we know will exist

Ozone depletion issues

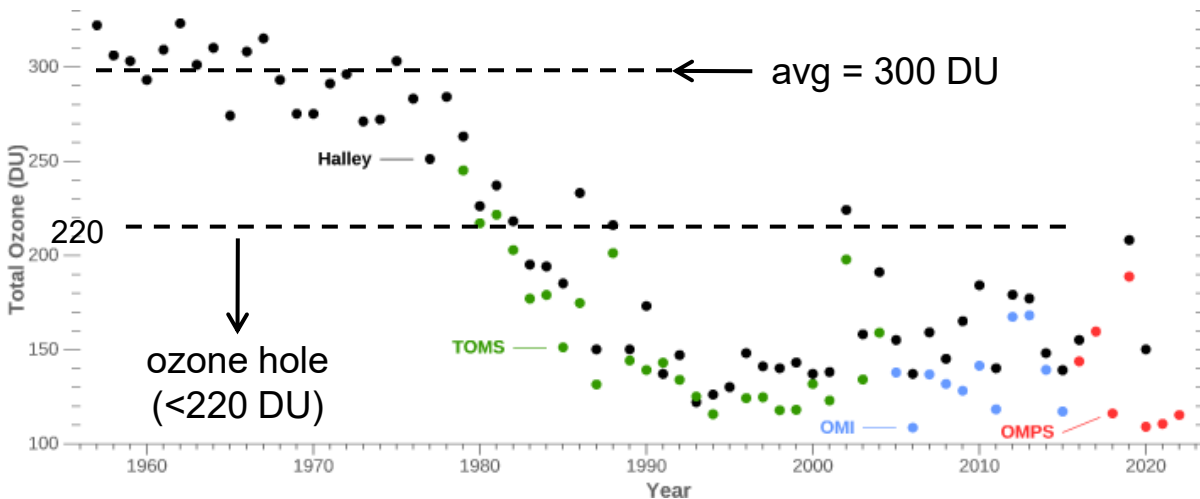
- Largest ozone reduction observed near N/S poles
- Substantial ozone reduction above Antarctic observed at the beginning of southern hemisphere spring
 - Antarctic ozone hole
 - Up to 70% reduction
 - When ozone column abundance < 220 DU: ozone hole
- Ozone reduction in Arctic also observed
 - Less severe (up to 25%)
 - Arctic ozone dent

What is the ozone hole?

The average amount of ozone in the atmosphere is roughly 300 Dobson Units (DU), equivalent to a layer 3 mm thick. Instruments on ground (at Halley) and above Antarctica (Total Ozone Mapping Spectrometer [TOMS] and Ozone Monitoring Instrument [OMI]) measured an acute drop in total atmospheric ozone during October in the early 1980s.

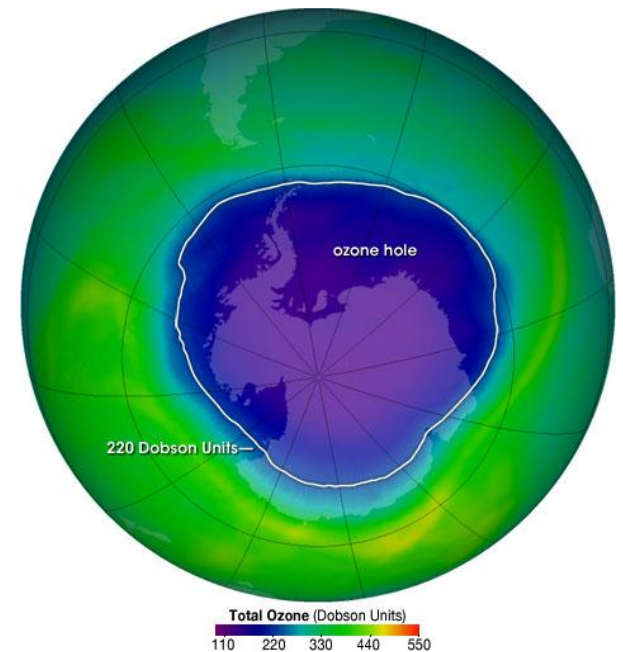
average ozone concentration (1 DU = 0.01 mm thickness)

October data; instruments located on ground at Halley and high above Antarctica (Total Ozone Mapping Spectrometer [TOMS] and Ozone Monitoring Instrument [OMI]).



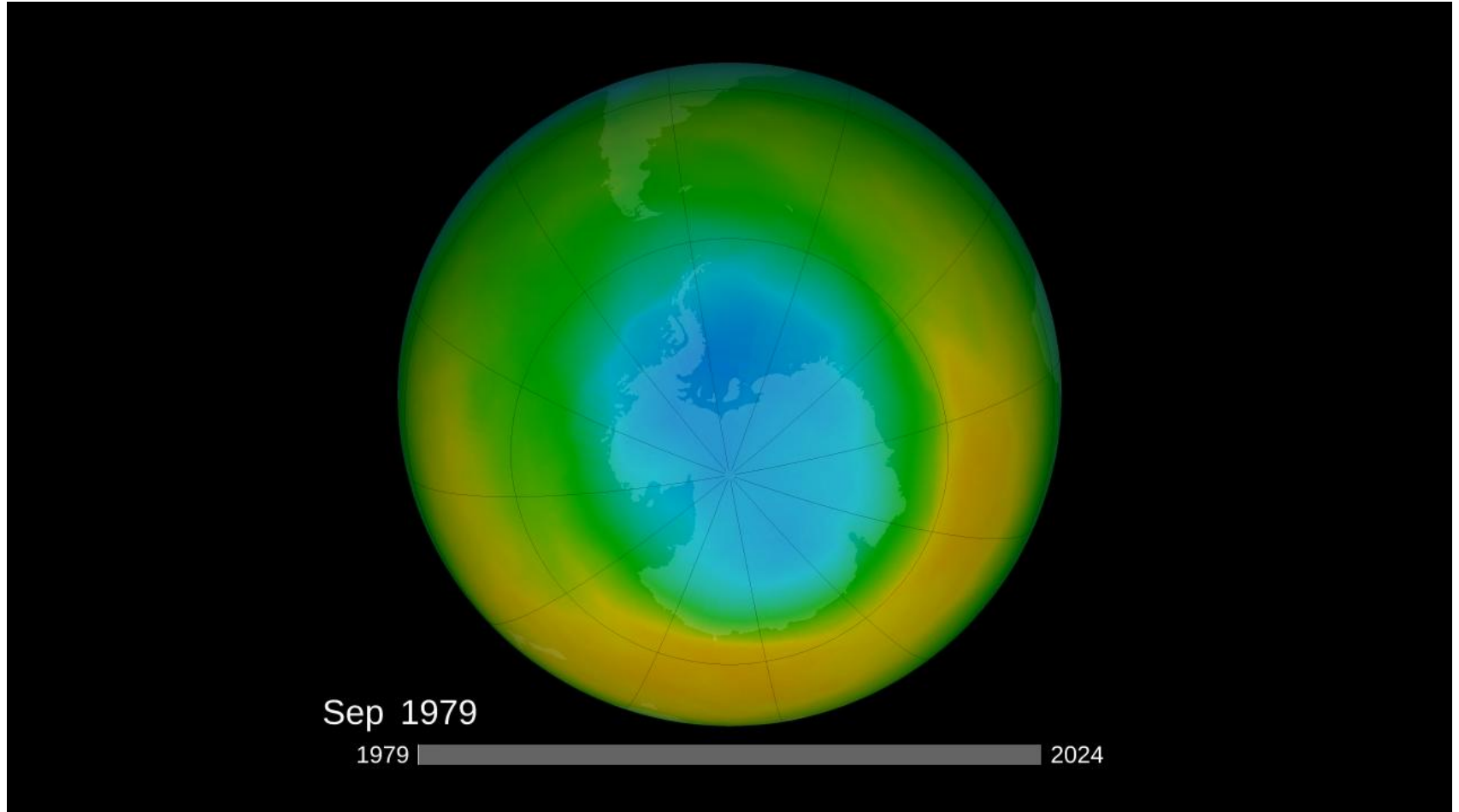
ozone hole area

The ozone “hole” is an area where ozone concentration drops to below 220 DU.



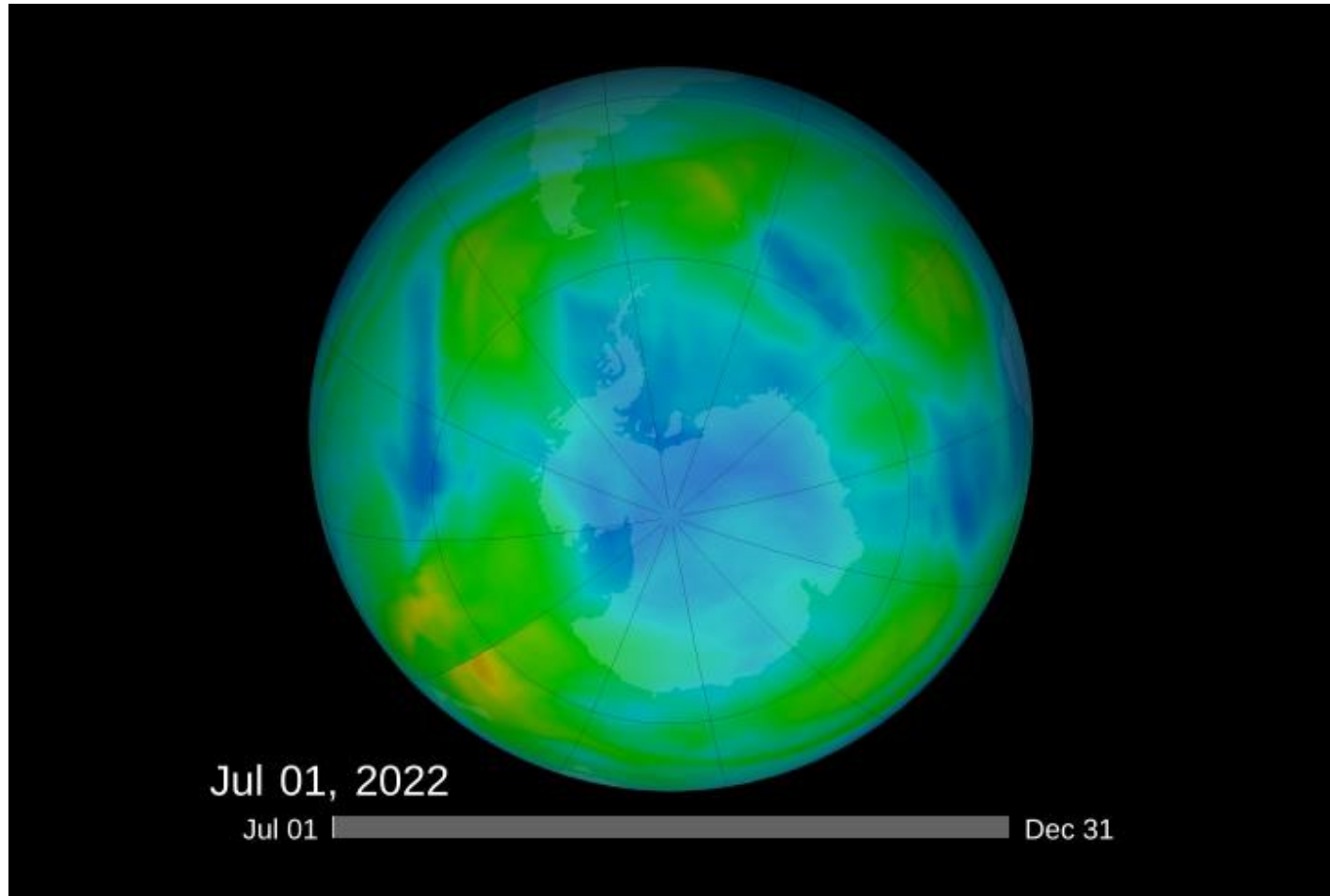
October 4, 2024

Ozone depletion issues



Antarctic ozone hole from 1979 to 2024
(NASA Ozone watch <http://ozonewatch.gsfc.nasa.gov/monthly/SH.html>)

Ozone depletion issues



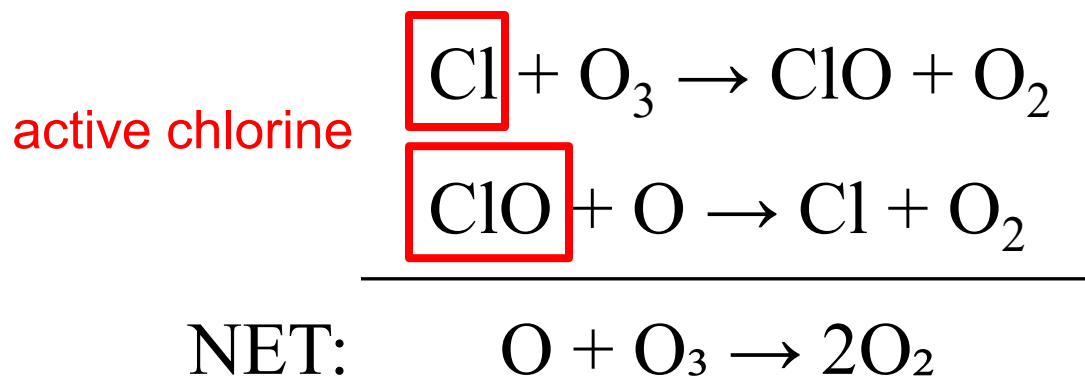
Antarctic ozone hole from 7/1/22 to 12/31/22
(NASA Ozone watch <http://ozonewatch.gsfc.nasa.gov/monthly/SH.html>)

Stratospheric ozone chemistry - CFCs

- Chlorofluorocarbons (CFC)
 - Contains chlorine (Cl) and fluorine (F)
 - Produced synthetically
 - Cheap, nontoxic, nonflammable, nonexplosive, insoluble, relatively inert
 - Used in refrigeration, spray cans, blowing agent
 - Average lifetime ~ 75 years
- There are also other halogenated species: HCFCs, HFCs, bromine containing chemicals...

Stratospheric ozone chemistry - CFCs

- In stratosphere, CFC break down when exposed to far-UV, release ClOx and atomic chlorine
- CFC don't break down in troposphere (why?)
 - Hence climb to stratosphere
- Destruct ozone through the generalized catalytic ozone destruction mechanism



Stratospheric ozone chemistry - CFCs

- -Cl chain length: 10 -1,000 (Jacobson, 2002)
 - One Cl molecule on average destructs 10-1,000 ozone molecules in lower and upper stratosphere, respectively. Aka chain length
 - Bigger chain length number, more powerful ozone destruction
- -Br chain length: 100 - 10,000 (Jacobson, 2002)
 - Slower reaction, more efficient
- The estimated numbers vary

Stratospheric ozone chemistry - CFCs

- Sink of active chlorine (Cl, ClO)
 - Reactions to form reservoir species: HCl and ClONO₂
 - Hydrochloric acid and chlorine nitrate
 - Temporarily store chlorine
 - When sunlight present, can photolyzed back to free chlorine, continue ozone destruction
- Bromine chemistry is similar

Ozone depletion issues

- Ozone reduction near N/S poles generally observed in/near spring time
 - Spring time differs for north/south hemisphere
- One factor: Cl and Br in stratosphere & their chemistry in polar stratospheric clouds (PSC)
 - Recall that active chlorine (Cl, ClO)/Br destroy ozone through generalized catalytic mechanism

Polar stratospheric clouds

Polar Stratospheric Cloud, seen from the NASA DC-8
on 14 January 2003 (photograph by Paul Newman,
GSFC) <http://remus.jpl.nasa.gov/kiruna/a015.htm>



Polar stratospheric clouds

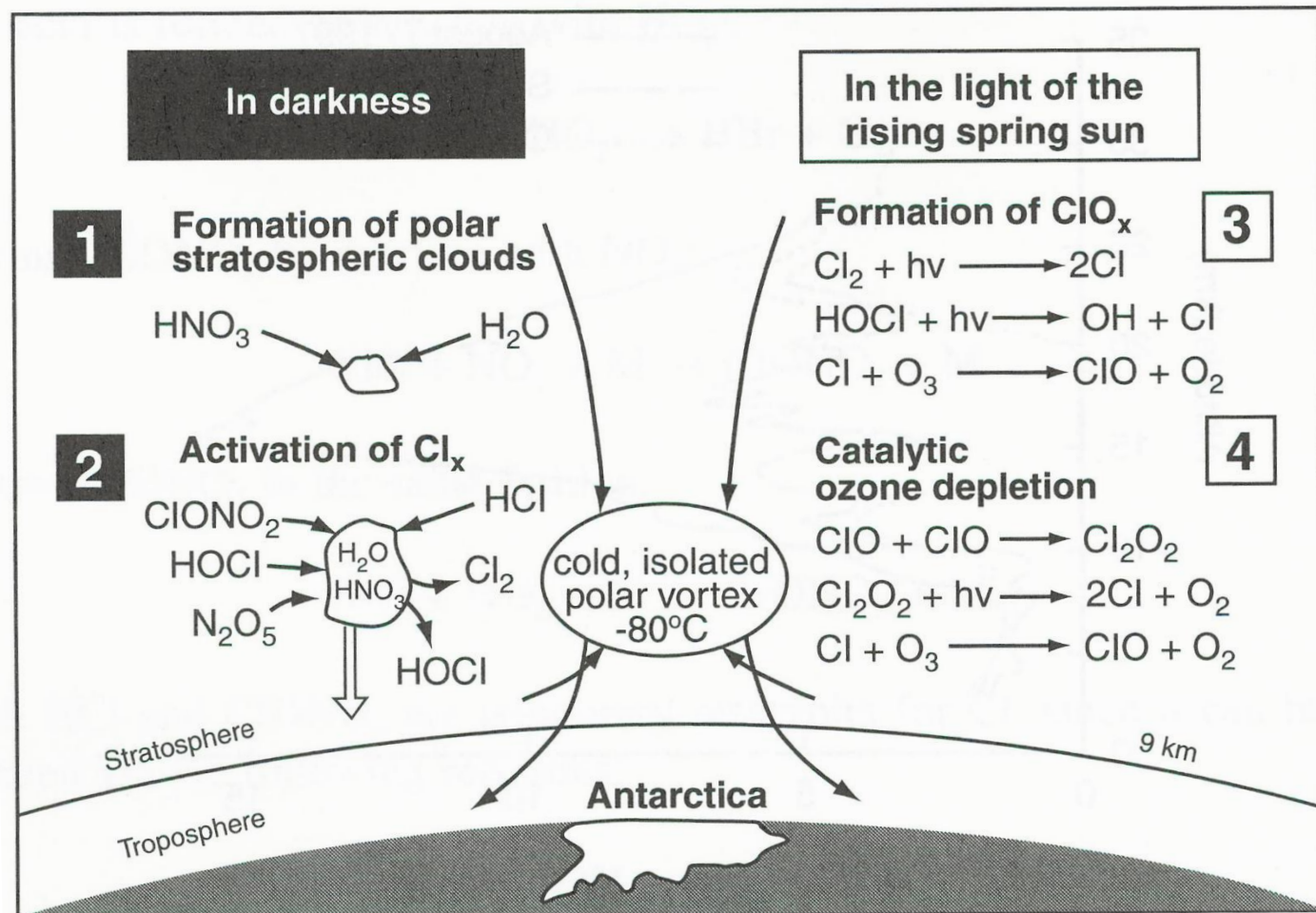


PSC as seen in Scandinavia (source unknown)

Ozone depletion process

- During polar night (before spring)
 - Polar stratospheric clouds (PSC) forms → reactions occur inside PSC → convert inactive Cl/Br (such as reservoir species HCl and ClONO₂) to species that are photo-chemically active (Cl₂, HOCl, ClNO₂ etc)
 - Chlorine activation
- Beginning of polar day (early spring)
 - Photo-chemically active species go through photolysis reaction → release active Cl/Br (Cl, ClO) → catalytically destroy ozone

Ozone depletion process



Ozone depletion process over Antarctica (Figure 4.11 Godish, 2003)

Ozone depletion process

- Why ozone depletion is less severe in north pole?
- Temperature higher in Arctic
 - Less PSC formation, less chlorine activation
 - Higher temperature due to difference in land cover
 - Ocean in Arctic vs land in Antarctic
 - Polar vortex not as strong in arctic due to rough terrain

Stratospheric Ozone Depletion

Websites:

<http://ozonewatch.gsfc.nasa.gov/>

<http://www.ozonelayer.noaa.gov/>

<http://www.youtube.com/watch?v=aKHcHmg4WKY>